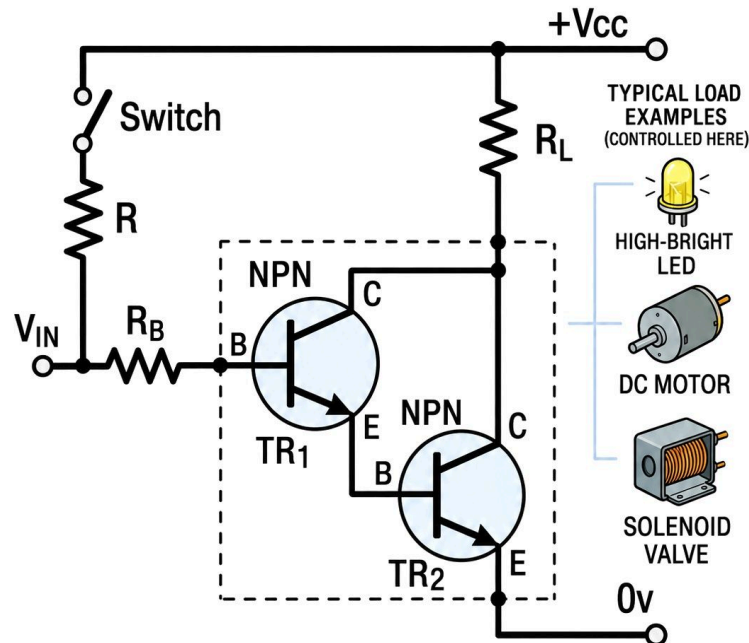




DARLINGTON SWITCH



$$\text{Overall Current Gain: } H_{FE_{Total}} \approx \beta_1 \cdot \beta_2$$

Combined transistors act as a massive-gain 'super-transistor', allowing a tiny base current to switch a very large load current.



Electronics Education

6 maj kl. 16:56 · 🌐

A Darlington switch uses two NPN transistors connected together to create very high current gain. The first transistor, TR1, amplifies the small input current, and the second transistor, TR2, amplifies it even more. Because of this, a tiny input signal can control a large load current.

When the switch is OFF, no base current flows, so both transistors remain OFF and the load does not operate. When the switch is ON, a small current flows into TR1 through resistor R_B. TR1 then drives TR2, allowing a much larger current to flow through the load.

This configuration acts like a powerful "super transistor" and is commonly used for driving LEDs, motors, relays, and solenoid valves from low-power control circuits such as microcontrollers and sensors.

The main advantage of a Darlington pair is its very high current amplification, making it ideal for switching heavy loads using very small input currents.

#ElectronicsEducation #ElectronicsRD #DarlingtonPair #TransistorSwitch #Electronics Visa mindre

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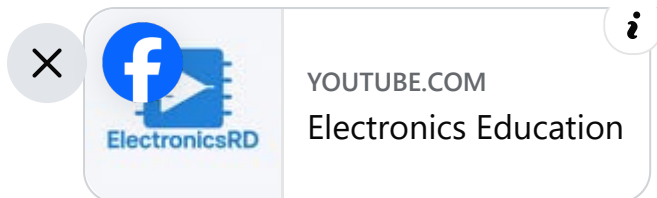
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Författare

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Youtube: https://youtube.com/@electronicseducation?sub_confirmation=1... Visa mer



1 v **Gilla** Svava 3



Mario Makraduli

You have switch and Vin, if Vin is LOW the swich can not activate the transistors!

1 v **Gilla** Svava 2

Visa 1 svar



Joel Petrique

A resistor between the base of TR1 and 0V is essential. Otherwise, given the high gain of this circuit, the transistor will pi... **Visa mer**

1 v **Gilla** Svava 1



Roderic Kenneth Bickerton

A modern FET MAKES A MUCH MORE EFFICIENT SWITCH.

1 v **Gilla** Svava 1



Cerqueira Nt Joaquim Xavier

It's a very important Post... especially given today's powerful applications, such as authomatic data compression.

. ... **Visa mer**

1 v **Gilla** Svava 3



Aleksandr Dolinin

Where is resistor from B of TR1 to E of TR2?

1 v **Gilla** Svava 1

Visa fler kommentarer

6 av 11



Skriv en kommentar ...

